

## *CSCE 190: Computing in the Modern World*

### Class 5 – GitHub Pages – Example of *Resources* (HTML/CSS, Markdown) via GitHub

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**Carolinian Creed: “I will practice personal and academic integrity.”**

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# Outline

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- Introduction Section
  - Recap from Last Class
- Main Section
  - Web Sites, Web Pages
  - GitHub Pages – example of sharing, HTML/CSS and Markdown
- Conclusion Section
  - Ask Me Anything

# Introduction Section

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# Recap from Last Lecture - Class 4 + Lab

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We talked about

- Sharing Resources – Content Format
- HTML, CSS
- Markdown
- GitHub pages

# Activity 4 – HTML / CSS, Markdown

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- Objective A: learn HTML/ CSS
- Tasks
  - See HTML cheat sheet
  - Create HTML files with an editor (like Notepad, vi) or online tool
  - Learn about different constructs
- Objective B: learn Markdown
- Tasks
  - See Markdown cheat sheet
  - Create Markdown files with an editor (like Notepad, vi)
  - Learn about different constructs

# Main Section

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# Concept: Web Site, Web Page

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- An online site, identified by a URL, consisting of one or more web pages
- Pre-requisites for a web site
  - A web address: *commonly* called getting a domain address
  - Site content: web pages
- Web page can be created by a website editor
  - GitHub pages
  - Google sites
  - Online tools – Wix, GoDaddy, ...
  - Editors – **Both WYSIWYG and Typesetting choices**

# Editor Concept: Format and Content

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**WYSIWYG** – what you see is what you get

- Software that lets you edit content so it looks just like the final result (format)
- Pros
  - Easy to edit
- Cons
  - If format changes slightly, may spend a lot of time fixing it, even if the content is same

## Examples

- Word Processors: Programs like Microsoft Word or Google Docs.
- **Website Builders: Tools like Wix or WordPress where you drag and drop items.**
- Email Clients: Gmail or Outlook when you change text color or add links

# Editor Concept: Format and Content

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**Typesetting** – separate content from format

- Software that lets you edit content separately;
- Format can be changed separately as well
- Pros
  - Easy to convert format
  - Advanced styles can be used seamlessly
- Cons
  - Harder to use software

## Examples

- Adobe InDesign: The professional industry standard for books, magazines, and complex layouts.
- LaTeX: A code-based system ideal for heavy academic, mathematical, and technical documents.
- **HTML/ CSS: html for content, css for digital styling**

# Default file: index.html

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- At a web address, index.html (or index.htm) is the file that a browser looks by default
- Otherwise, the name of a file has to be explicitly mentioned as part of a URL call
  - Example: <https://biplav-s.github.io/> (file not mentioned; index.html is assumed)
- Question: what happens if file is mentioned ?
  - The named file is opened
  - Example: <https://github.com/biplav-s/biplav-s.github.io/blob/master/index.html>

# Default file: Readme.md

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- On GitHub, the file that is looked by default is Readme.md
- Example: <https://github.com/biplav-s/biplav-s.github.io/>
  - File opened: Readme.md

# Creating One's GitHub Page

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- Have a GitHub account, say ExampleAccount
- A website for you is automatically provided: ExampleAccount.github.io
- Default location for website content is provided:
  - ExampleAccount.github.io/ExampleAccount/ExampleAccount.github.io
  - Create a sample index.html

# Sample index.html

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```
<!DOCTYPE html>
<html>
<head>
  <title>CSCE 190: [your name]</title>
</head>

<body>
  <h1> [your name] </h1>
  <h3> [your name] – Hello World!</h3>
</body>
</html>
```

Content of index.html

Adapted from: Robert Bailey's presentation

# Example 2: TA 1's Website

A simple Fun Fact website using Github Pages

Link: [https://apallavias04-source.github.io/DemoPage\\_class190/](https://apallavias04-source.github.io/DemoPage_class190/)

apallavias04-source.github.io/DemoPage\_class190/



# HTML elements used

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**<!DOCTYPE html>** — Declaration (not a tag) that tells the browser to render the page using modern HTML5 rules.

**<html>** — The root element that wraps the entire page.

**<head>** — Holds metadata about the page that isn't shown directly on screen.

**<meta>** — Supplies metadata like character encoding and viewport settings; here it sets UTF-8 encoding and makes the page mobile-responsive.

**<title>** — Sets the text shown in the browser tab; here it's "Welcome."

**<style>** — Contains the page's CSS; here it defines the background image, card styling, and button look.

**<body>** — Holds everything visible on the page.

**<div>** — A generic container with no inherent meaning; used twice here - once for the glass .card wrapper, once for the clickable #box.

**<h1>** — The top-level heading; displays "Welcome."

**<script>** — Contains the page's JavaScript; here it holds the facts array and the **showFact()** function that swaps in a random fact on click.

# Example 3: TA 2's Website

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A personal academic portfolio website hosted on GitHub Pages with a custom domain

Link: <https://aataiwo.com>

Source Code: <https://github.com/Akian-coder/portfolio>



# HTML elements used

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`<!DOCTYPE html>` — Tells the browser to use modern HTML5.

`<html>` — The main container for the entire website.

`<head>` — Contains page information, fonts, styling, and the website title.

`<meta>` — Sets character encoding, mobile responsiveness, and site description.

`<style>` — Contains the CSS used for the dark theme, navigation bar, responsive layout, cards, and buttons.

`<body>` — Contains all visible website content.

`<header>` and `<nav>` — Create the top navigation menu for Home, About, Skills, Research, Projects, and Contact.

`<section>` — Organizes the website into different content areas.

`<a>` — Creates links to sections of the website and external platforms such as GitHub and LinkedIn.

`<img>` — Displays images, including the profile image and project images.

# Discussion – Audience, Content, Elements

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- Audience: decide who will see the website / webpage
- Content: decide what content to put
- Page Elements:
  - Simple is better. Have a reason to include a feature
  - Keep it mobile friendly
  - Beyond content, focus on web page loading performance
  - Keep evolving over time

# Assignment 5

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- Assignment
- Create a Github account
- Create a github.io repository on Github
- Connect to your repo and Commit a change
- Using VS Code, the Github website, Github dev, etc.
- Test your github.io link

# Concluding Section

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# Conclusion

We talked about

- Web Sites, Web Pages
- GitHub Pages – example of sharing, HTML/CSS and Markdown

| Week | Content                                             | Assignment                       |
|------|-----------------------------------------------------|----------------------------------|
| 1    | Intro, syllabus, university resources               | Slack account                    |
| 2    | CSE majors, career paths, research areas            | Write a mission statement        |
| 3    | Github (account and basics) + Lab                   | Github account, repo             |
| 4    | HTML basics, IDE (VS Code)                          | None                             |
| 5    | More HTML, CSS, Problem Solving                     | Update Github with CSS           |
| 6    | Brainstorming, problem statement                    | Problem statement                |
| 7    | Computing – (Software) Dev. Life Cycle              | Draft Requirement, Specification |
| 8    | Computing – Testing a System                        | Test Plan                        |
| 9    | AI – LLMs/FMs, Generative AI, Issues and Challenges | Problem Solving with GenAI       |
| 10   | Building a resume, LinkedIn                         | Draft resume in PDF              |
| 11   | Review resumes                                      | Review resumes, update Github    |
| 12   | Convert resume to HTML                              | Begin conversion, update Github  |
| 13   | CSE research areas                                  | (Finalize website)               |
| N/A  | Last day of classes: 4 Dec, 2026                    | <b>All extensions end</b>        |

# Ask Me Anything

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